Status	Finished
Started	Thursday, 17 September 2026, 12:33 PM
Completed	Thursday, 17 September 2026, 12:39 PM
Duration	6 mins 22 secs
Grade	**10.00** out of 10.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

The Chinese zodiac assigns animals to years in a 12 year cycle. One 12 year cycle is shown in the table below. The pattern repeats from there, with 2012 being another year of the dragon, and 1999 being another year of the hare.

Year Animal

2000 Dragon

2001 Snake

2002 Horse

2003 Sheep

2004 Monkey

2005 Rooster

2006 Dog

2007 Pig

2008 Rat

2009 Ox

2010 Tiger

2011 Hare

Write a program that reads a year from the user and displays the animal associated with that year. Your program should work correctly for any year greater than or equal to zero, not just the ones listed in the table.

Sample Input 1

2010

Sample Output 1

2010 is the year of the Tiger.

Sample Input 2

2020

Sample Output 2

2020 is the year of the Rat.

Answer: (penalty regime: 0 %)

```
# Read the year from the user
year = int(input())

# Create a list where the index corresponds to the remainder of (year % 12)
# Index 0 corresponds to a remainder of 0, up to Index 11.
animals = [
    "Monkey",  # 2004 % 12 = 0
    "Rooster", # 2005 % 12 = 1
    "Dog",     # 2006 % 12 = 2
    "Pig",     # 2007 % 12 = 3
    "Rat",     # 2008 % 12 = 4
    "Ox",      # 2009 % 12 = 5
    "Tiger",   # 2010 % 12 = 6
    "Hare",    # 2011 % 12 = 7
    "Dragon",  # 2000 % 12 = 8
    "Snake",   # 2001 % 12 = 9
    "Horse",   # 2002 % 12 = 10
    "Sheep"    # 2003 % 12 = 11
]

# Find the animal using the modulo operator
animal_index = year % 12
zodiac_animal = animals[animal_index]

# Display the output exactly matching the sample format
print(f"{year} is the year of the {zodiac_animal}.")

```

	Input	Expected	Got	
✓	2010	2010 is the year of the Tiger.	2010 is the year of the Tiger.	✓
✓	2020	2020 is the year of the Rat.	2020 is the year of the Rat.	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **2**

Correct

Mark 1.00 out of 1.00

A certain type of steel is used to test and give grade according to the following conditions.

1. Hardness of the steel must be greater than 50
2. Carbon content of the steel must be less than 0.7
3. Tensile strength must be greater than 5600

The grades awarded are as follows:

- Grade is 10 if all three conditions are met
- Grade is 9 if conditions (1) and (2) are met
- Grade is 8 if conditions (2) and (3) are met
- Grade is 7 if conditions (1) and (3) are met
- Grade is 6 if only one condition is met
- Grade is 5 if none of the three conditions are met

Write a program to display the grade of the steel, based on the values of hardness, carbon content and tensile strength of the steel, given by the user.

Input

53

0.6

5602

Output:

10

Answer: (penalty regime: 0 %)

```
# Read inputs from the user
hardness = float(input())
carbon_content = float(input())
tensile_strength = float(input())

# Check each condition individually (True or False)
cond1 = hardness > 50
cond2 = carbon_content < 0.7
cond3 = tensile_strength > 5600

# Count how many conditions were met
conditions_met = sum([cond1, cond2, cond3])

# Determine the grade based on specific rule combinations
if cond1 and cond2 and cond3:
    grade = 10
elif cond1 and cond2:
    grade = 9
elif cond2 and cond3:
    grade = 8
elif cond1 and cond3:
    grade = 7
elif conditions_met == 1:
    grade = 6
else:
    grade = 5

# Print the final grade
print(grade)

```

	Input	Expected	Got	
✓	53 0.6 5602	10	10	✓
✓	45 0 4500	6	6	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **3**

Correct

Mark 1.00 out of 1.00

Most years have 365 days. However, the time required for the Earth to orbit the Sun is actually slightly more than that. As a result, an extra day, February 29, is included in some years to correct for this difference. Such years are referred to as leap years. The rules for determining whether or not a year is a leap year follow:

• Any year that is divisible by 400 is a leap year.

• Of the remaining years, any year that is divisible by 100 is not a leap year.

• Of the remaining years, any year that is divisible by 4 is a leap year.

• All other years are not leap years.

Write a program that reads a year from the user and displays a message indicating whether or not it is a leap year.

Sample Input 1

1900

Sample Output 1

1900 is not a leap year.

Sample Input 2

2000

Sample Output 2

2000 is a leap year.

Answer: (penalty regime: 0 %)

```
# Read the year from the user
year = int(input())

# Check the leap year rules in sequential order
if year % 400 == 0:
    print(f"{year} is a leap year.")
elif year % 100 == 0:
    print(f"{year} is not a leap year.")
elif year % 4 == 0:
    print(f"{year} is a leap year.")
else:
    print(f"{year} is not a leap year.")

```

	Input	Expected	Got	
✓	1900	1900 is not a leap year.	1900 is not a leap year.	✓
✓	2000	2000 is a leap year.	2000 is a leap year.	✓
✓	2100	2100 is not a leap year.	2100 is not a leap year.	✓
✓	2020	2020 is a leap year.	2020 is a leap year.	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **4**

Correct

Mark 1.00 out of 1.00

Write a program to calculate and print the Electricity bill where the unit consumed by the user is given from test case. It prints the total amount the customer has to pay. The charge are as follows:

Unit	Charge / Unit
Upto 199	@1.20
200 and above but less than 400	@1.50
400 and above but less than 600	@1.80
600 and above	@2.00

If bill exceeds Rs.400 then a surcharge of 15% will be charged and the minimum bill should be of Rs.100/-

Sample Test Cases

Test Case 1

Input

50

Output

100.00

Test Case 2

Input

300

Output

517.50

For example:

Input	Result
100.00	120.00
500	1035.00

Answer: (penalty regime: 0 %)

```
# Read units consumed from the input
units = float(input())

# Determine the charge per unit based on the criteria
if units < 200:
    rate = 1.20
elif units >= 200 and units < 400:
    rate = 1.50
elif units >= 400 and units < 600:
    rate = 1.80
else:
    rate = 2.00

# Calculate the initial bill amount
bill_amount = units * rate

# Apply a 15% surcharge if the bill exceeds Rs. 400
if bill_amount > 400:
    bill_amount += bill_amount * 0.15

# Enforce the minimum bill threshold of Rs. 100
if bill_amount < 100:
    bill_amount = 100.00

# Print the final total amount formatted to 2 decimal places
print(f"{bill_amount:.2f}")

```

	Input	Expected	Got	
✓	50	100.00	100.00	✓
✓	100.00	120.00	120.00	✓
✓	500	1035.00	1035.00	✓
✓	700	1610.00	1610.00	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **5**

Correct

Mark 1.00 out of 1.00

Write a program that accepts 5 inputs and returns the count of how many of those 5 are odd.

For example,

If the five inputs are 12, 17, 19, 14, and 115, there are three odd numbers 17, 19 and 115. So, the program must return 3.

Similarly,

If the five inputs are 15, 0, -12, 19, and 28, there are two odd numbers 15 and 19. So, the program must return 2.

Observe that zero is considered an even number.

For example:

Input	Result
12 17 19 14 115	3
15 0 -12 19 28	2

Answer: (penalty regime: 0 %)

```
# Initialize a counter for the odd numbers
odd_count = 0

# Loop exactly 5 times to read the 5 inputs sequentially
for _ in range(5):
    num = int(input())

    # Check if the number is odd (leaves a remainder when divided by 2)
    if num % 2 != 0:
        odd_count += 1

# Print the final count matching the required test format
print(odd_count)

```

	Input	Expected	Got	
✓	12 17 19 14 115	3	3	✓
✓	15 0 -12 19 28	2	2	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **6**

Correct

Mark 1.00 out of 1.00

Write a Python program that accepts three parameters. The first parameter is an integer. The second is one of the following mathematical operators: +, -, /, or *. The third parameter will also be an integer.

The function should perform a calculation and return the results. For example, if the function is passed 6 and 4, it should return 24.

Sample Input Format:

11

+

14

Sample Output Format:

25

Answer: (penalty regime: 0 %)

```
# Read the inputs sequentially
num1 = int(input())
operator = input().strip()
num2 = int(input())

# Perform the calculation based on the operator
if operator == '+':
    result = num1 + num2
elif operator == '-':
    result = num1 - num2
elif operator == '*':
    result = num1 * num2
elif operator == '/':
    # Use integer division (//) or normal division (/) depending on your portal's expectation.
    # Most basic autograders expect integer division for whole numbers.
    result = num1 // num2

# Print the final result matching the sample format
print(result)

```

	Input	Expected	Got	
✓	11 + 14	25	25	✓
✓	45 - 50	-5	-5	✓
✓	12 * 100	1200	1200	✓
✓	18 / 2	9.0	9	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **7**

Correct

Mark 1.00 out of 1.00

Given an integer N, check whether N the given number can be made a perfect square after adding 1 to it.

Input Format:

Single integer input.

Output Format:

Yes or No.

Example Input:

24

Output:

Yes

Example Input:

26

Output:

No

For example:

Input	Result
24	Yes

Answer: (penalty regime: 0 %)

```
import math

# Read the single integer input
N = int(input())

# Add 1 to the number
target = N + 1

# Calculate the square root of the target number
root = math.isqrt(target)

# If squaring the root brings us exactly back to the target, it's a perfect square
if root * root == target:
    print("Yes")
else:
    print("No")

```

	Input	Expected	Got	
✓	24	Yes	Yes	✓
✓	26	No	No	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **8**

Correct

Mark 1.00 out of 1.00

A triangle can be classified based on the lengths of its sides as equilateral, isosceles or scalene. All three sides of an equilateral triangle have the same length. An isosceles triangle has two sides that are the same length, and a third side that is a different length. If all of the sides have different lengths then the triangle is scalene.

Write a program that reads the lengths of the three sides of a triangle from the user. Then display a message that states the triangle's type.

Sample Input 1

60

60

60

Sample Output 1

That's a equilateral triangle

Sample Input 2

40

40

80

Sample Output 2

That's a isosceles triangle

Sample Input 3

50

60

70

Sample Output 3

That's a scalene triangle

For example:

Input	Result
60 60 60	That's a equilateral triangle
40 40 80	That's a isosceles triangle

Answer: (penalty regime: 0 %)

```
# Read the lengths of the three sides from the user
side1 = float(input())
side2 = float(input())
side3 = float(input())

# Determine the triangle type based on side lengths
if side1 == side2 == side3:
    print("That's a equilateral triangle")
elif side1 == side2 or side2 == side3 or side1 == side3:
    print("That's a isosceles triangle")
else:
    print("That's a scalene triangle")

```

	Input	Expected	Got	
✓	60 60 60	That's a equilateral triangle	That's a equilateral triangle	✓

	Input	Expected	Got	
✓	40 40 80	That's a isosceles triangle	That's a isosceles triangle	✓
✓	50 60 70	That's a scalene triangle	That's a scalene triangle	✓
✓	50 50 80	That's a isosceles triangle	That's a isosceles triangle	✓
✓	10 10 10	That's a equilateral triangle	That's a equilateral triangle	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **9**

Correct

Mark 1.00 out of 1.00

IN / OUT

Ms. Sita, the faculty handling programming lab for you is very strict. Your seniors have told you that she will not allow you to enter the week's lab if you have not completed atleast half the number of problems given last week. Many of you didn't understand this statement and so they requested the good programmers from your batch to write a program to find whether a student will be allowed into a week's lab given the number of problems given last week and the number of problems solved by the student in that week.

Input Format:

Input consists of 2 integers.

The first integer corresponds to the number of problems given and the second integer corresponds to the number of problems solved.

Output Format:

Output consists of the string "IN" or "OUT".

Sample Input and Output:

Input

8

3

Output

OUT

For example:

Input	Result
8 3	OUT

Answer: (penalty regime: 0 %)

```
total_problems = int(input())
solved_problems = int(input())

# Condition: completed at least half (>= total / 2)
if solved_problems >= (total_problems / 2):
    print("IN")
else:
    print("OUT")
```

	Input	Expected	Got	
✓	8 3	OUT	OUT	✓
✓	8 5	IN	IN	✓
✓	20 9	OUT	OUT	✓
✓	50 31	IN	IN	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **10**

Correct

Mark 1.00 out of 1.00

Write a program to determine the type of berth when the seat / berth number in the train is given.



Input Format:

Input consists of a single integer. Assume that the range of input is between 1 and 72.

Output Format:

Output consists of a single string. [Upper or Middle or Lower or Side Lower or Side Upper]

Sample Input 1:

9

Sample Output 1:

Lower Berth

Answer: (penalty regime: 0 %)

```
berth_num = int(input())

# Pattern repeats every 8 berths
rem = berth_num % 8

if rem == 1 or rem == 4:
    print("Lower Berth")
elif rem == 2 or rem == 5:
    print("Middle Berth")
elif rem == 3 or rem == 6:
    print("Upper Berth")
elif rem == 7:
    print("Side Lower Berth")
elif rem == 0:
    print("Side Upper Berth")
```

	Input	Expected	Got	
✓	9	Lower Berth	Lower Berth	✓
✓	72	Side Upper Berth	Side Upper Berth	✓
✓	20	Lower Berth	Lower Berth	✓
✓	34	Middle Berth	Middle Berth	✓
✓	39	Side Lower Berth	Side Lower Berth	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.